CHAPTER 15

The Autonomic Nervous System

Involuntary control of the body's organs and internal balance.

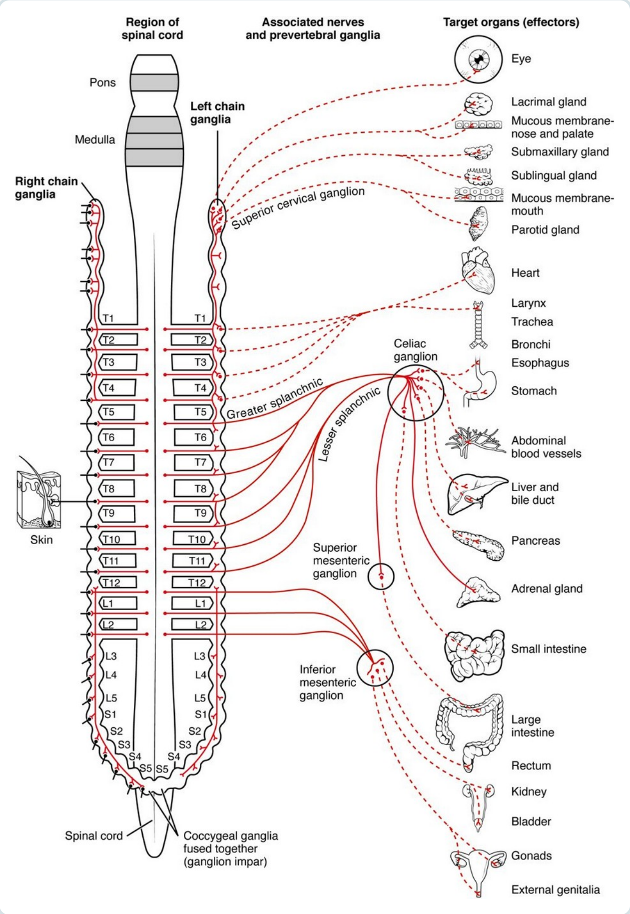
Anatomy & Physiology · Broward-Miami Health Institute

FACULTY EDITION

Learning Objectives

After studying this chapter, students will be able to:

1. Describe the components of the autonomic nervous system
2. Differentiate between the structures of the sympathetic and parasympathetic divisions in the autonomic nervous system
3. Name the components of a visceral reflex specific to the autonomic division to which it belongs
4. Predict the response of a target effector to autonomic input on the basis of the released signaling molecule



The autonomic nervous system: involuntary control.

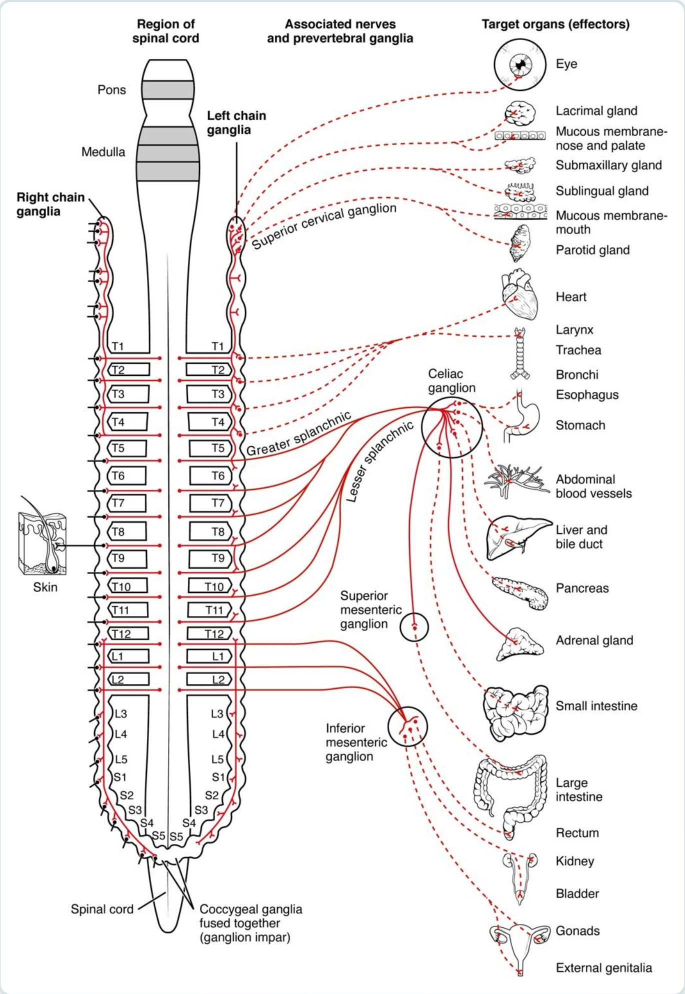
15.1

SECTION

Divisions of the Autonomic Nervous System

The sympathetic and parasympathetic divisions of the autonomic system.

The Sympathetic Division

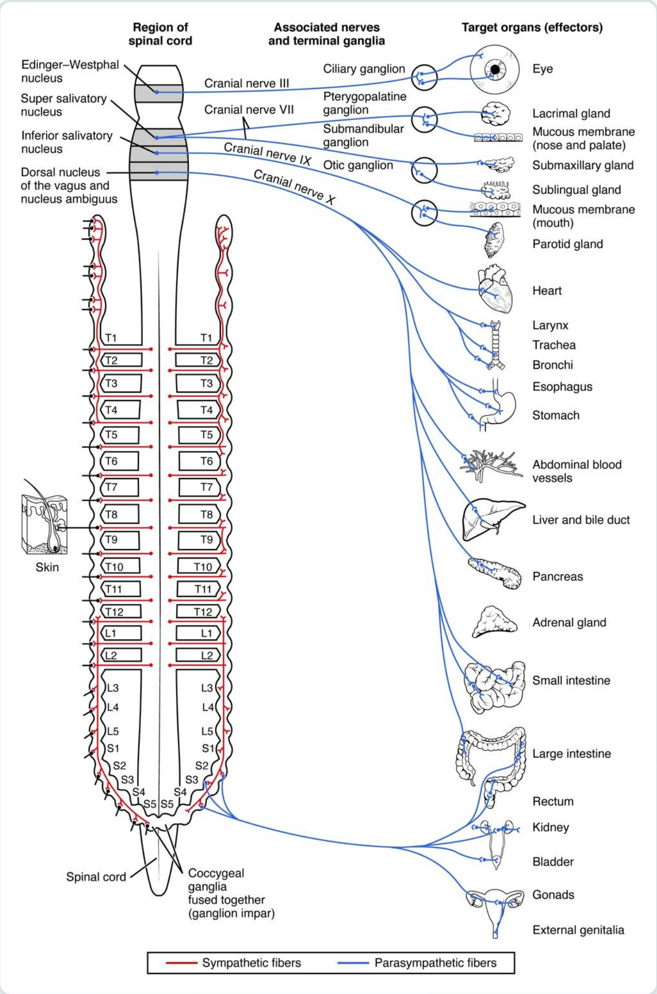


The sympathetic division

- The "fight-or-flight" system
- Prepares the body for action
- Raises heart rate and breathing
- Arises from the thoracolumbar cord

SECTION 15.1

The Parasympathetic Division

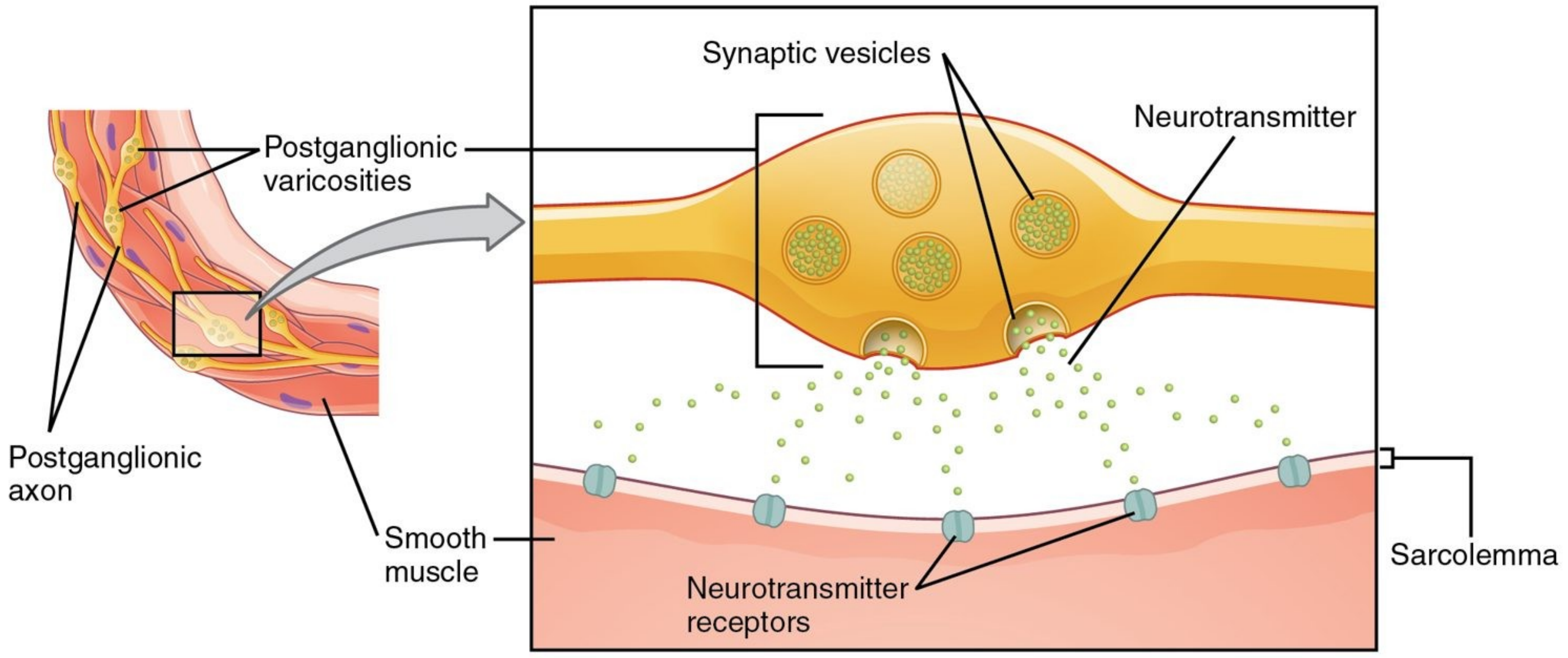


The parasympathetic division

- The "rest-and-digest" system
- Conserves energy at rest
- Slows the heart, aids digestion
- Arises from cranial and sacral regions

SECTION 15.1

Autonomic Neurotransmitters



Neurotransmitters of the autonomic system

- A two-neuron pathway to the target
- Acetylcholine and norepinephrine
- Receptor type determines the effect
- The basis of many drug actions

15.2

SECTION

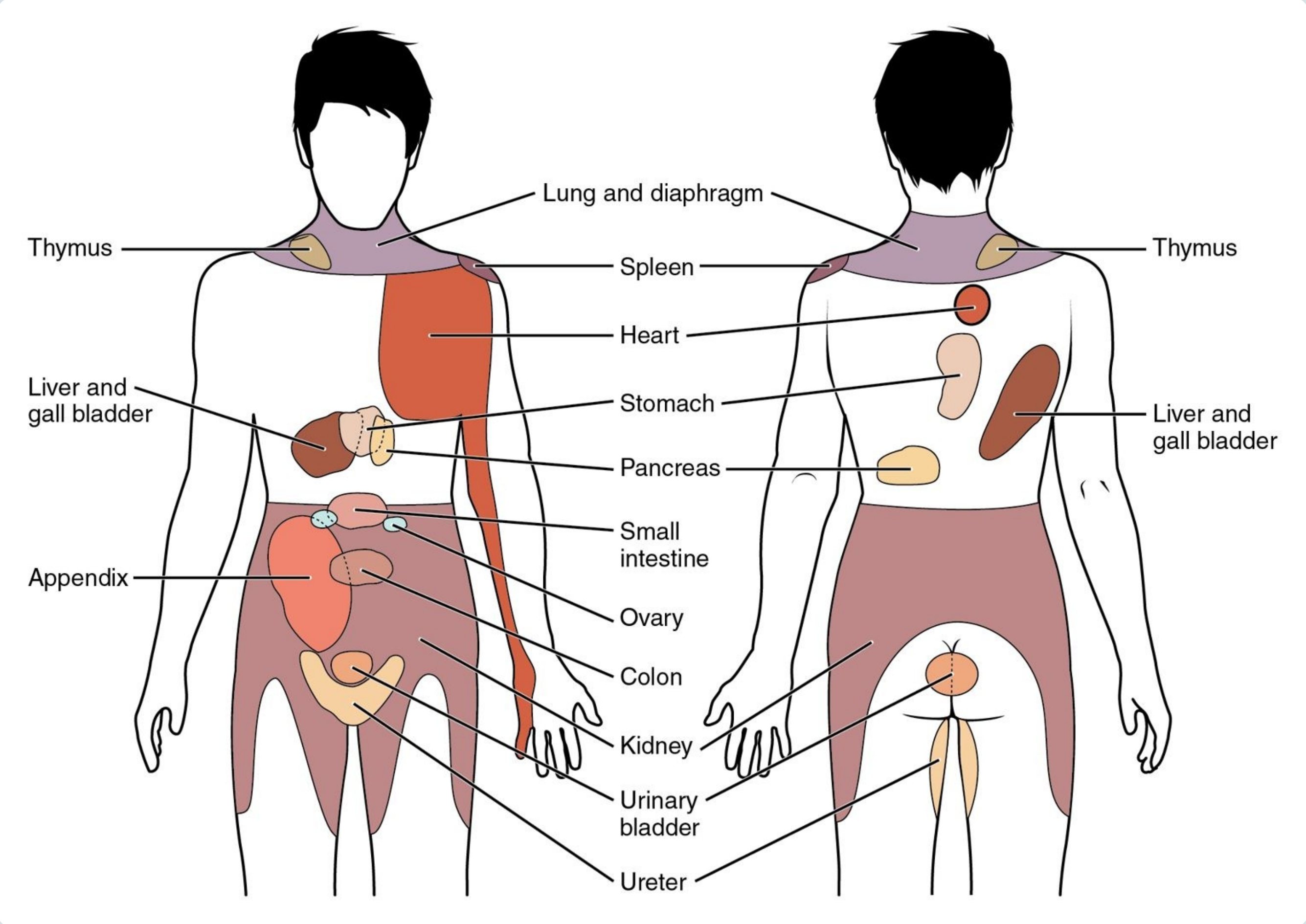
Autonomic Reflexes and Homeostasis

How autonomic reflexes keep internal conditions stable.

SECTION 15.2

Dual Innervation

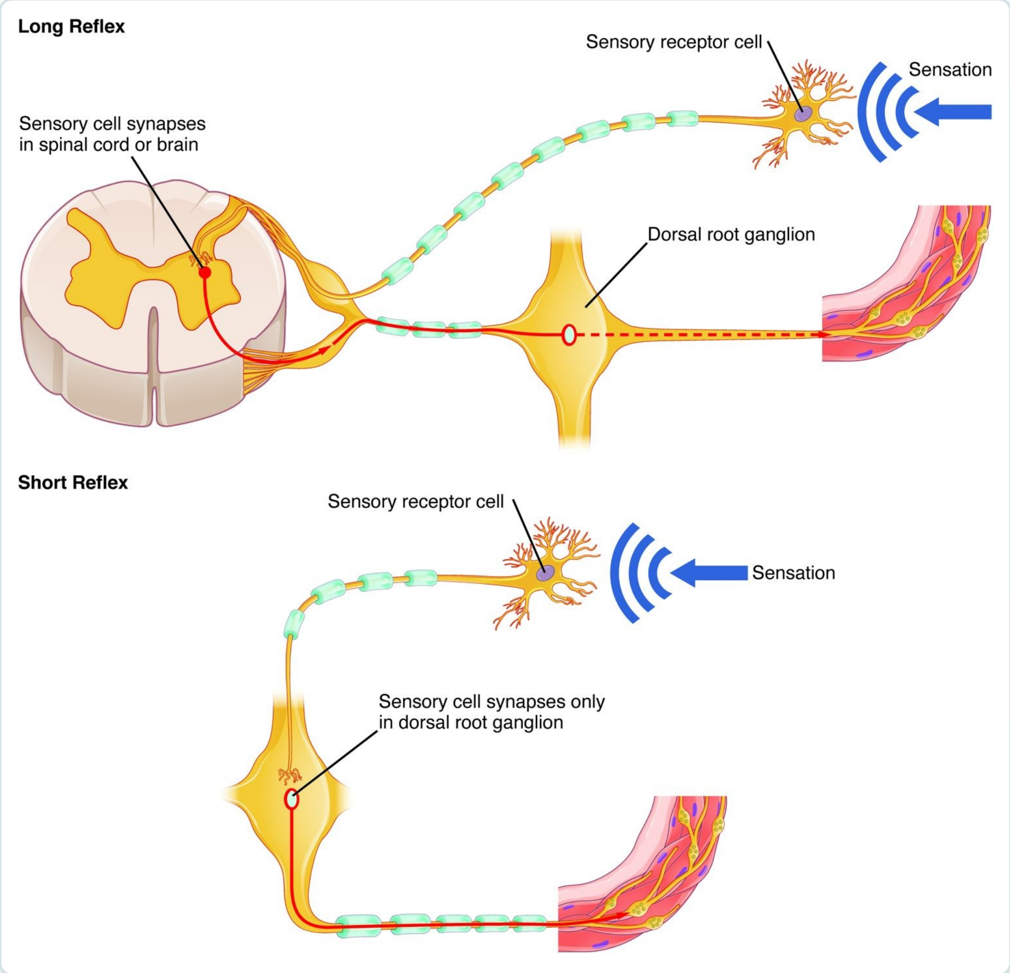
- Most organs get both divisions
- The two usually oppose each other
- Balance fine-tunes organ activity
- Maintains homeostasis



Sympathetic and parasympathetic effects on organs

Autonomic Reflexes

- Automatic responses to stimuli
- Sensory input → CNS → autonomic output
- Control heart rate, BP, digestion
- Operate without awareness



Long and short autonomic reflexes

15.3

SECTION

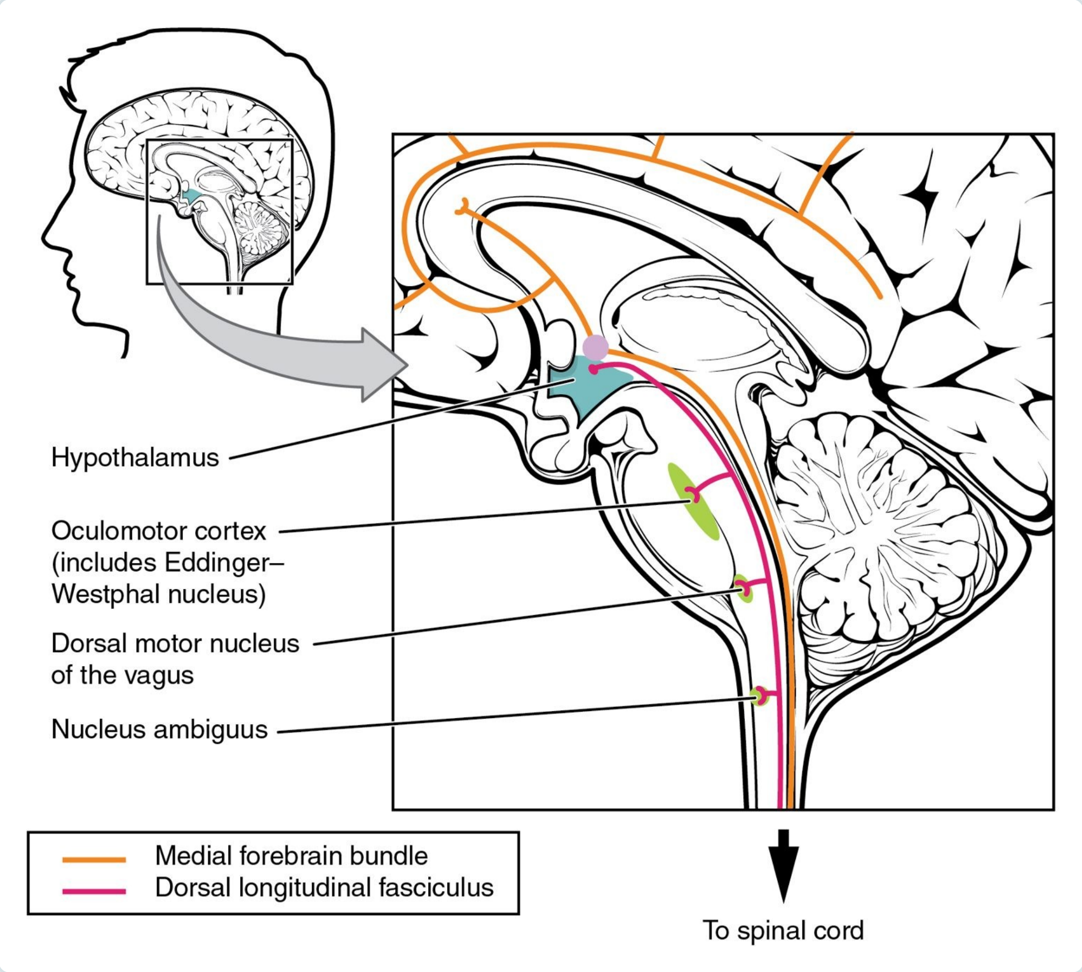
Central Control

How the brain oversees the autonomic nervous system.

SECTION 15.3

Central Control

- The hypothalamus is the main controller
- Integrates autonomic and endocrine output
- Brainstem centers run vital reflexes
- Linked to emotion and stress



Central control of the autonomic system

15.4

SECTION

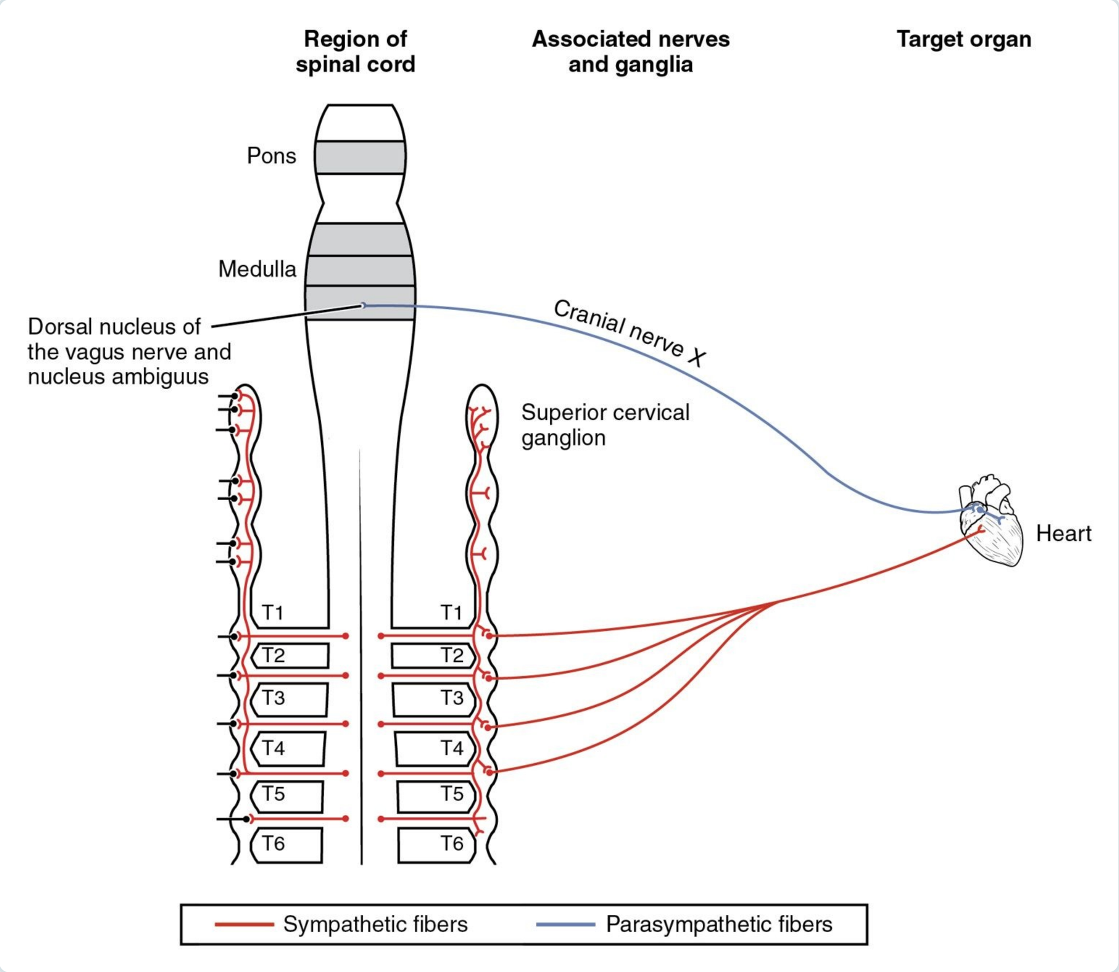
Drugs that Affect the Autonomic System

How drugs act on the autonomic system to treat disease.

SECTION 15.4

Autonomic Drug Targets

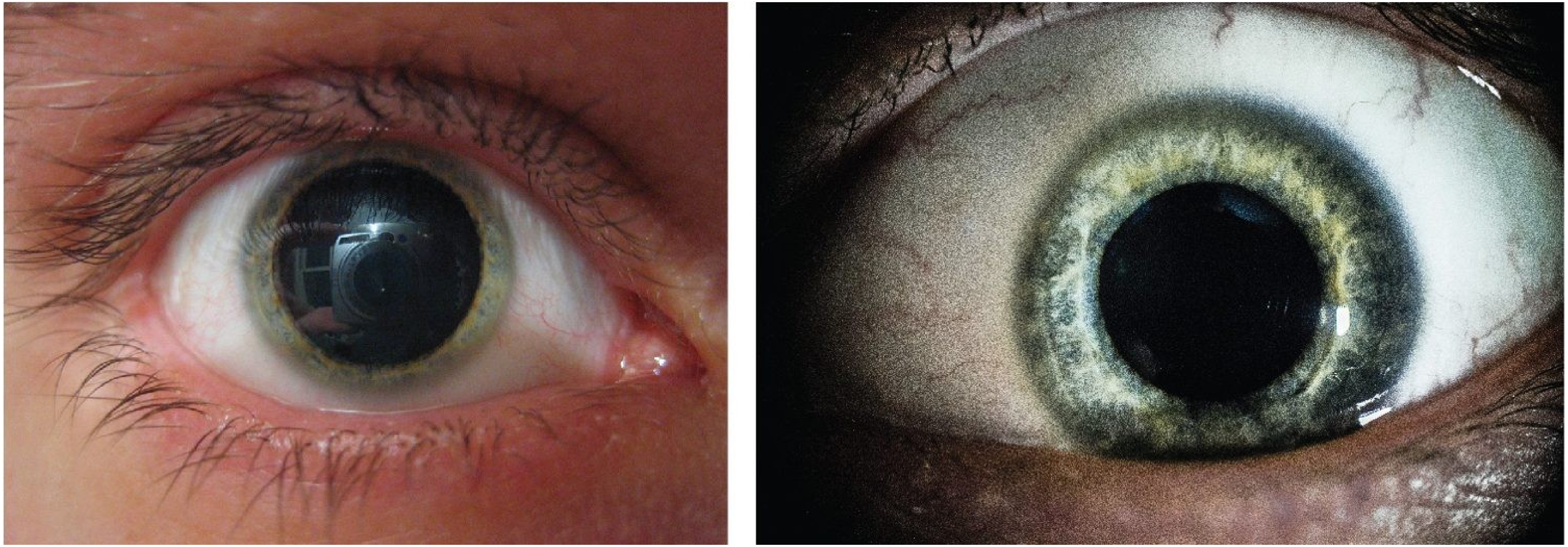
- Drugs mimic or block autonomic signals
- Affect the heart, vessels, lungs, eyes
- Beta-blockers slow the heart
- Many common medications act here



Autonomic control of the heart — a drug target

SECTION 15.4

Drug Effects on the Pupil



Autonomic drug effects on the pupil

- Autonomic drugs change pupil size
- Sympathetic action dilates (mydriasis)
- Parasympathetic action constricts (miosis)
- Used in eye exams and diagnosis

CHAPTER 15

Key Takeaways

- The autonomic nervous system controls involuntary functions of organs, glands, and vessels.
- The sympathetic division drives "fight-or-flight"; the parasympathetic handles "rest-and-digest."
- A two-neuron pathway uses acetylcholine and norepinephrine, with receptor type setting the effect.
- Most organs receive both divisions, whose balance maintains homeostasis.
- The hypothalamus and brainstem control the system, a major target for drugs.

CLINICAL CONNECTIONS

Why the Autonomic System Matters in Practical Nursing

Blood Pressure

Autonomic reflexes regulate BP and are altered by many drugs.

Beta-Blockers

Block sympathetic signals to slow the heart and lower BP.

Fight-or-Flight

The stress response explains symptoms of anxiety and shock.

Pupil Assessment

Pupil size and reactivity reveal autonomic and brain function.

Bladder & Bowel

Autonomic control underlies continence and digestion.

Atropine

Blocks parasympathetic effects to speed the heart and dry secretions.